



Gestió de Bases de Dades

UD 2 - Entity / Relationship Model

1r CFGS Desenvolupament d'Aplicacions Web
Teacher: Toni de Andrés





BLOCK 1 – DATABASE ENGINEERING

LESSON 1 – Introduction to databases

LESSON 2 – Logical Design: Entity-Relationship Model

LESSON 3 – Logical to Physical Design: Relational Model

BLOCK 2 – DATABASE MANAGEMENT

LESSON 4 – Physical Design: Data Definition Language

LESSON 5 – Implementation Stage: Data Management Language

BLOCK 3 – DATABASE MAINTENANCE

LESSON 6 – Implementation Stage: SQL Embedded in a Formal Language

LESSON 7 – Maintenance Stage: Triggers

LESSON 8 – Maintenance Stage: Redesigning the DB Model

LESSON 9 – Data Security Management



UD2 - Objectives

After completing this lesson, you should be able to:

- Place E/R Model in context of Modeling Data
- Identify
 - On one hand **Entities**, their **attributes**
 - On the other hand **Relationships** between entities
- from a case of study
- Diagram all of them

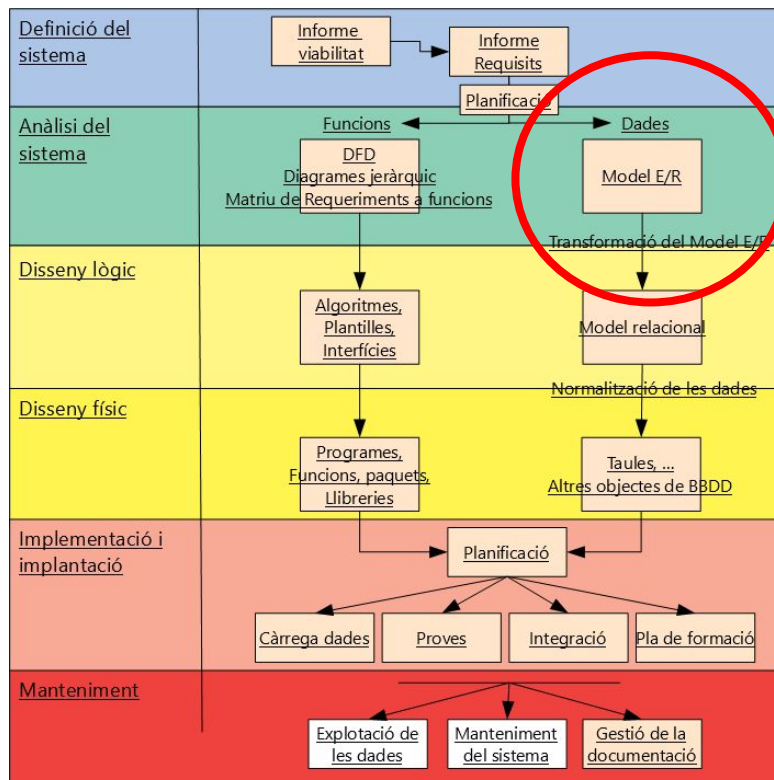


UD2 - E/R Model

1. Introduction to E/R Diagram
 - a. [Introduction](#)
 - b. [What does an ERD provide ?](#)
2. Components of a E/R Diagram
 - a. [Entities](#)
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UD2 - Introduction to E/R Diagram - Introduction





UD2 - Introduction to E/R Diagram

ERD provides:

- ◉ A simple understanding way to explain how data is organized.
 - ◉ A picture that users and computer scientist can easily understand how data is represented
 - ◉ A clear and precise way to show data requirements for an organization.
 - ◉ A way to easily develop and refine the data model.
 - ◉ A clear picture of the scope of the project.
 - ◉ A framework for integrating multiple applications, development projects, and purchased application packages.
-
- ◉ Therefore it is called Universe of Discourse at the most real approach that the data engineer makes about the piece of universe which is intended to model -not the whole universe.

UD2 - Introduction to E/R Diagram

First of all, it is highly relevant to get a clear and complete vision and a good knowledge of the system (and the organization or enterprise).



Lack of experience in diagramming is often a problem to perform this task.



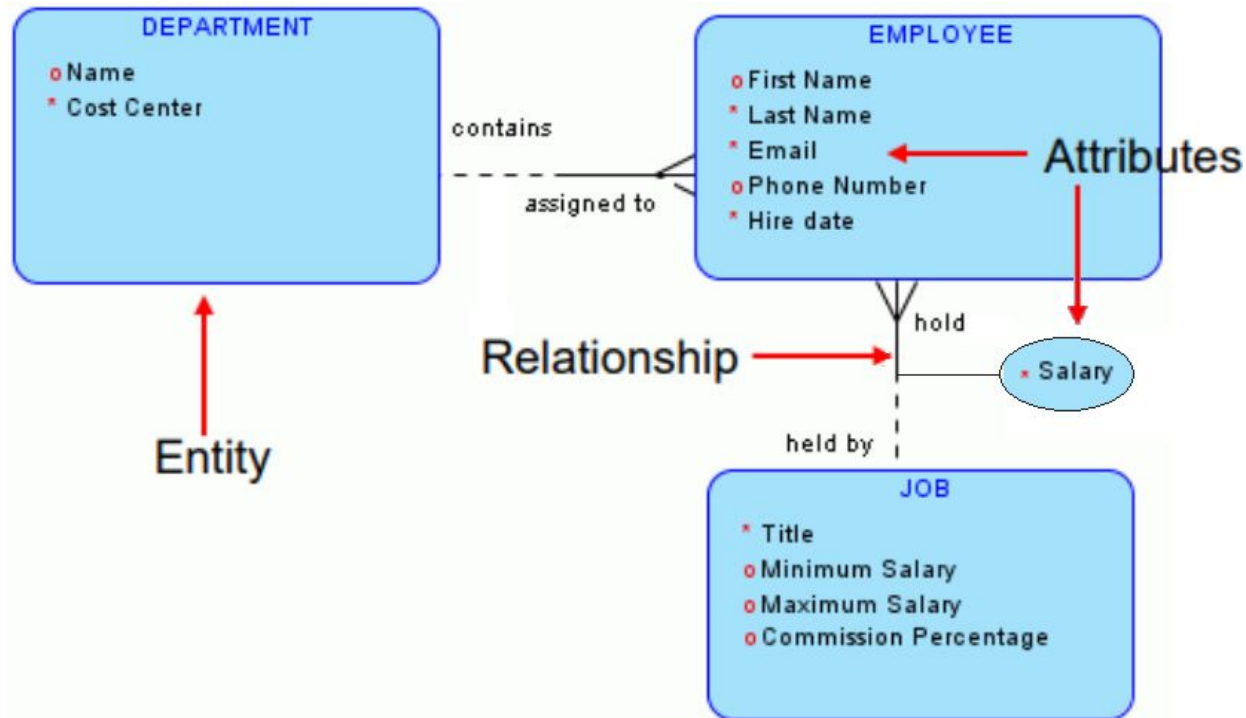
UD2 - Introduction to E/R Diagram

The diagram for a logical data model is called an **Entity/Relationship Diagram (ERD)** which is a high level (and semantic) data model to translate the meanings and interactions of real-world of an enterprise into a conceptual scheme:

- ERD is the most used system to represent a Relational Data Model.
- Represents a collection of basic objects (entities) we got from the perception of a real world (**Universe of Discourse**), their attributes which give us a completely description of each one and lastly the relationships among these objects. All together are expressed graphically by an ERD.



UD2 - Components of a E/R Diagram





UD2 - Components of a E/R Diagram - Entities

An entity is ...

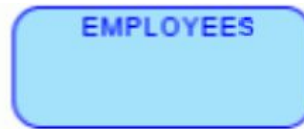
- A "container" for things -items- that it is wanted to store.
- An Instance (or item) of an Entity represents one element from the real world we want to store.
- Therefore, an Entity represents a set of items grouped by the same characteristics on a Universe of Discourse.



UD2 - Components of a E/R Diagram - Entities

An entity is represented by ...

- ... rectangles, the entity name is written inside the rectangle (in capital letters).
- MUST be named as a **noun** (singular or plural, but the same criteria for the ERD).

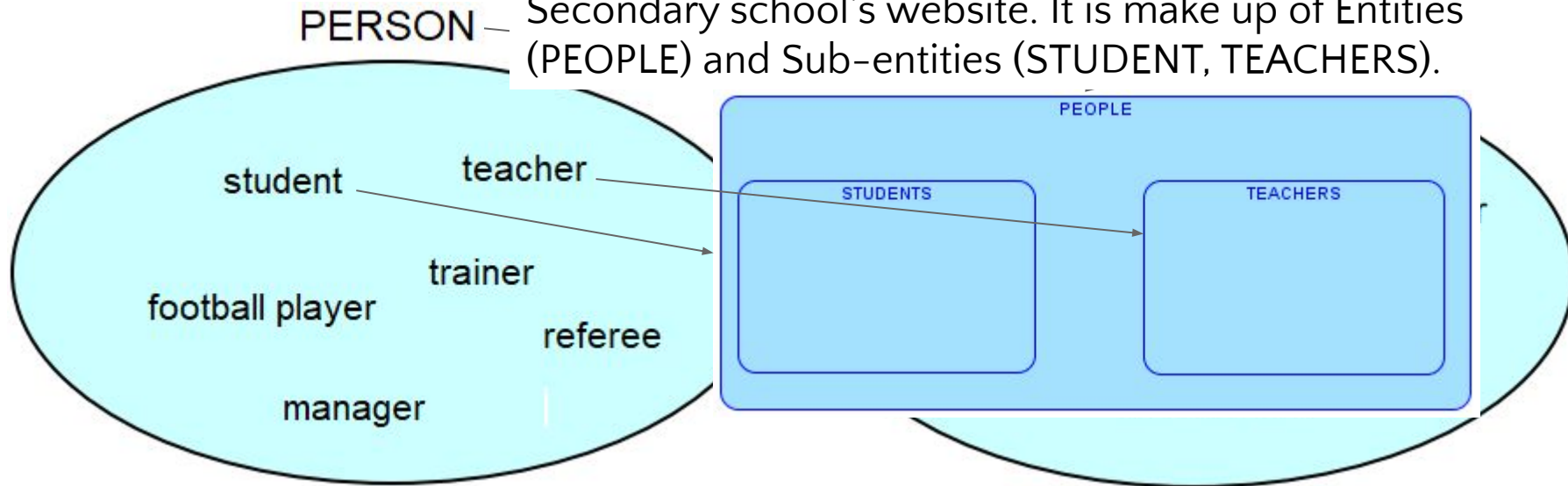




UD2 - Components of a E/R Diagram - Entities

An Entity can represent a set of instances that are of interest to a particular business.

Suppose this typical case of study: an ERD for a Secondary school's website. It is make up of Entities (PEOPLE) and Sub-entities (STUDENT, TEACHERS).





UD2 - Components of a E/R Diagram - Entities

Entities are a compound of items I can put together because of their relationship between them. Below some examples ...

- Related to People or Companies:
 - Students, Teachers, Staff, ...
 - Players, Referees, Managers ...
 - Customers, Suppliers ...
- Related to Places:
 - Secondary Schools, Classes, ...
 - Stadiums, Dressing rooms, ...
 - Departments, Areas, Zones, ...
 - Countries, States, Provinces, Cities or Villages, Districts, Streets ...
- Related to Intangible items:
 - Categories of something (*Type of ...*), ...
 - Header of something (e.g. *Invoices* or *Budgets* as a set of services or articles) ...
 - A combination of something (e.g. *Shifts* as a combination of days and working times) ...



UD2 - Components of a E/R Diagram - Attributes

Attributes are ...

- Each one of data that describe every item of an entity.
- The specific data is wanted to be stored for an item.
- The name of the attribute should not include the name of the entity. because attributes describe the entity.
- MUST be named as a **singular noun**.
 - EXAMPLE: '*surname1*' or '*surname2*' instead of '*surnames*'



UD2 - Components of a E/R Diagram - Attributes

Key Attributes are ...

- ... identifying easily an instance (item) in an entity.
- Items of an Entity can be identified by one or more attributes.

Examples:

- The number of a player in his/her back is identifying each player in a football team. This attribute is called as **Key Attribute**.
- The NSS for an Employee is identifying every one of them.
- Sometimes an item may be identified by a sequenced numeration: 1, 2, 3, ... so it is said: 'the item number one, the number two, ...'.



UD2 - Components of a E/R Diagram - Attributes

Optional attributes are used when ...

- the value of an attribute for a certain entity is unknown,
- does not have any applicable value for the attribute at this moment,

Mandatory attributes are used when ...

- the value of an attribute for a certain entity is known and we want to store a value for this attribute.

Examples:

- *phone-number*: unknown value at this moment
- *rent-data-from*: the value is known but *rent-data-to* is unknown at this moment



UD2 - Components of a E/R Diagram - Attributes

Attributes are often ...

- ... defining a plain feature of an item. This feature is also known as atomic value. Otherwise, we can work within:
 - Multivalued attributes,
 - Derived (or calculated) attributes,
 - Repetitive attributes.



UD2 - Components of a E/R Diagram - Attributes

Multivalued attributes ...

... has a set of values that can be listed separately. Every single value is often separated by a specified character, e.g. ' ' (blank space) or ',' (comma).

Examples:

- '*complete-name*': formed by:
 - first name, surname1, surname2
- '*address*': consisting of :
 - address (street, number, floor, door), location, state, postal code, country



UD2 - Components of a E/R Diagram - Attributes

Derived (or calculated) attributes are ...

The value for this type of attribute can be derived from the values of other related attributes or entities.

Example:

- suppose that a student entity has an attribute named '*age*', which indicates the student's age. If the student entity also has an attribute '*date-of-birth*', we can calculate age from date-of-birth and the current date. Thus, '*age*' is a derived attribute. Somehow, take into account that '*date-of-birth*' can provide more information than '*age*' (e.g. horoscope)



UD2 - Components of a E/R Diagram - Attributes

Repetitive attributes are ...

A limited number of attributes representing a plain single attribute.

Examples:

- phone-number: anybody may have zero, one or several phone numbers: Phone-Num1, Phone-Num2, Phone-Num3, ...
- movie-produced-by: a film co-produced by many producers: Producer1, Producer2, Producer3, ...
- A list of skills for a Person: Skill1, Skill2, Skill3, ...



UD2 - Components of a E/R Diagram - Attributes

Multivalued, Derived and Repetitive attributes must ...

... not be present at the end of Designing phase, hence should be cleared up from Physical Design.

This question will be treated later.



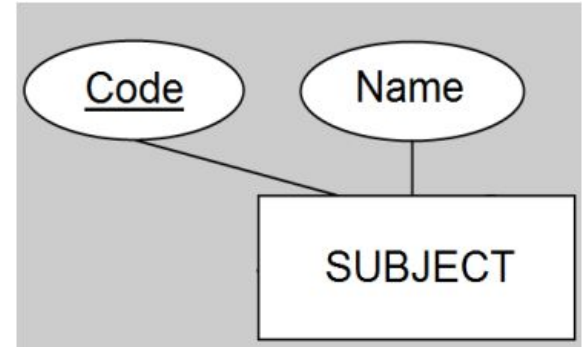
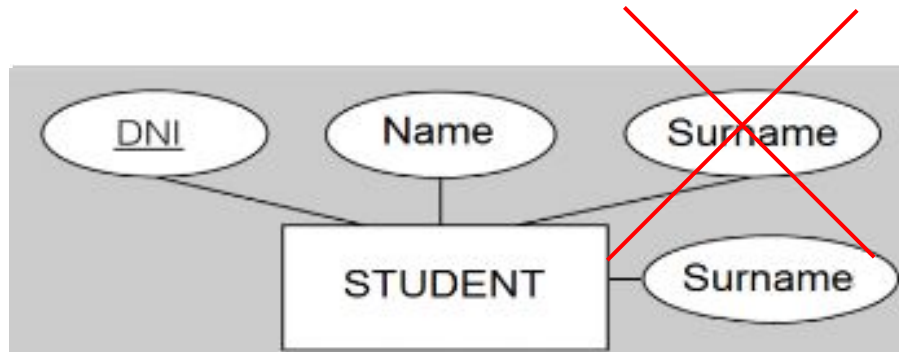
UD2 - Components of a E/R Diagram - Attributes

There are many ways to represent attributes, e.g. ...

... an ellipse in lower letters and a line (solid for mandatory attribute, and broken for non) which link attributes to entity.

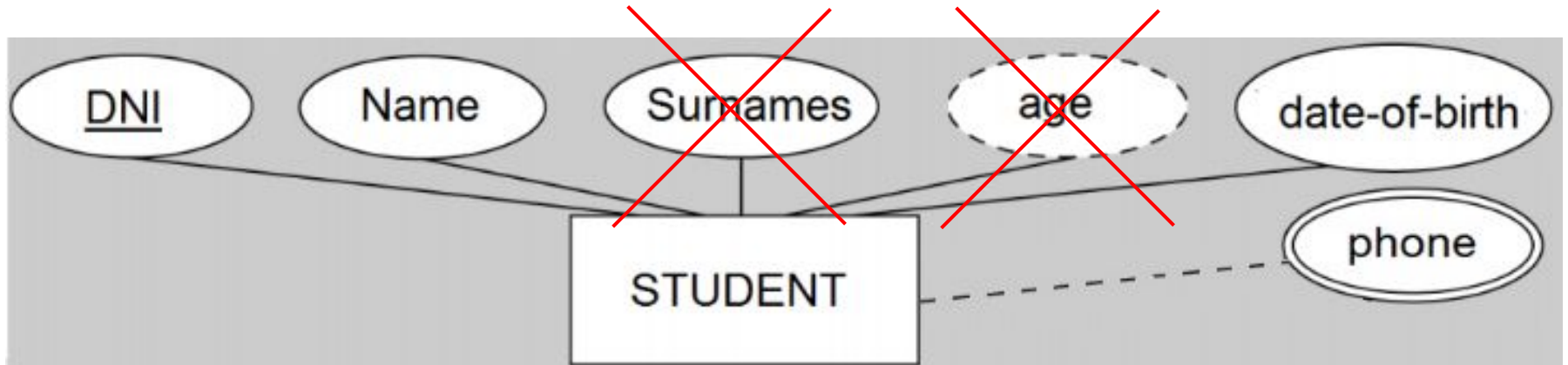
- **Key attributes** are represented by underline.
- **Multivalued and Repetitive attributes** are represented by double ellipses.
- **Derived attributes** are represented by dashed ellipses.
- **Optional attributes** are shown with a broken line linked to its entity.

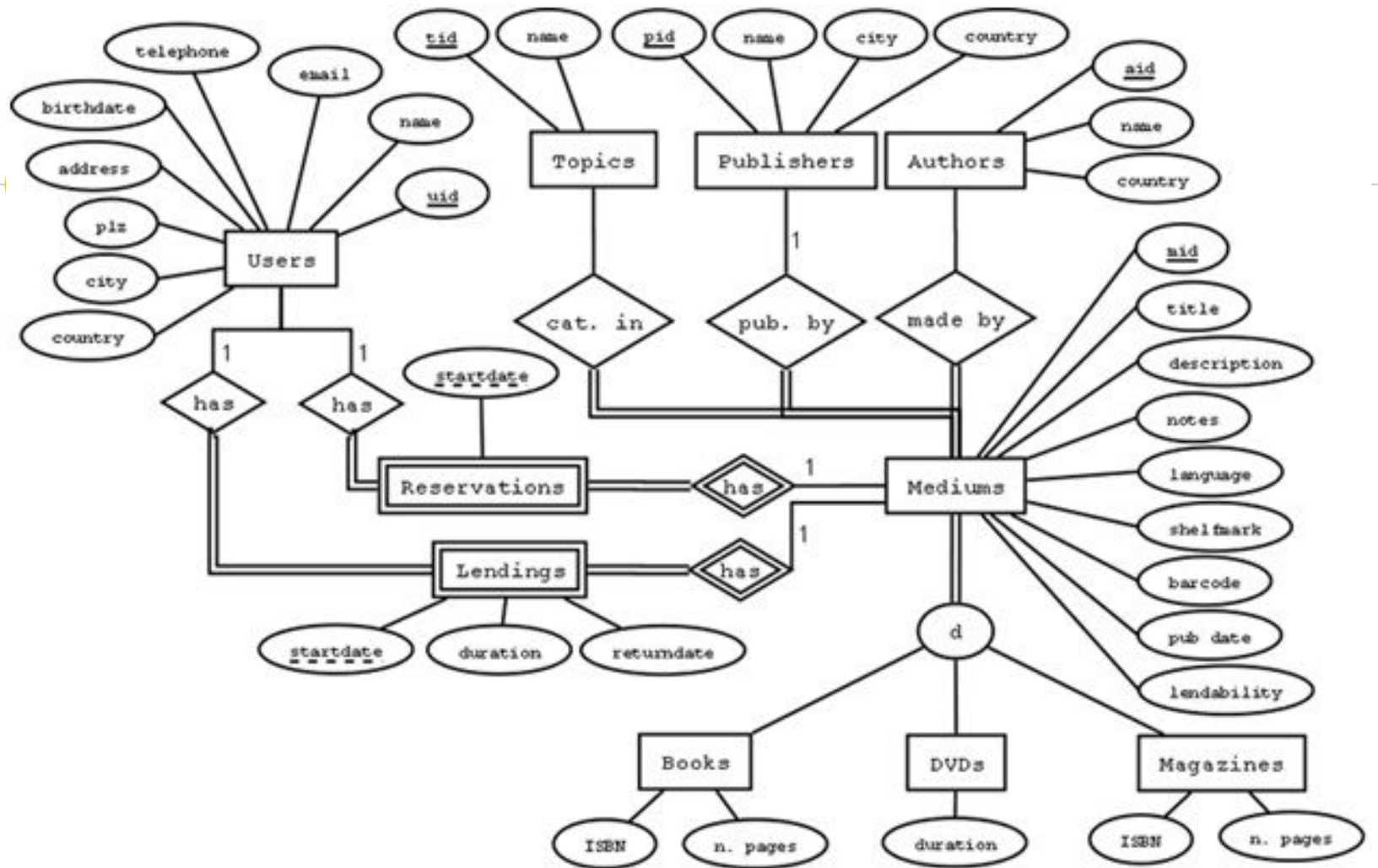
UD2 - Components of a E/R Diagram - Attributes





UD2 - Components of a E/R Diagram - Attributes







UD2 - Components of a E/R Diagram - Attributes

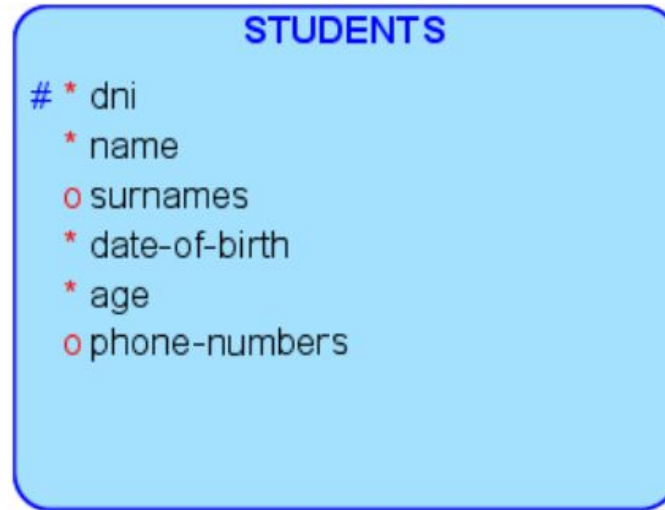
Attributes are also represented by ...

Attributes can be shown within the entity box.

- ◉ Attributes are either:
 - **Key Attribute**, indicated by a #
 - **Mandatory**, indicated by *
 - **Optional**, indicated by o
 - **Multivalued, Derived and Repetitive** attributes do not have a specific representation

UD2 - Components of a E/R Diagram - Attributes

The notation we will use (Barker Notation) ...





UD2 - Components of a E/R Diagram - Relationships

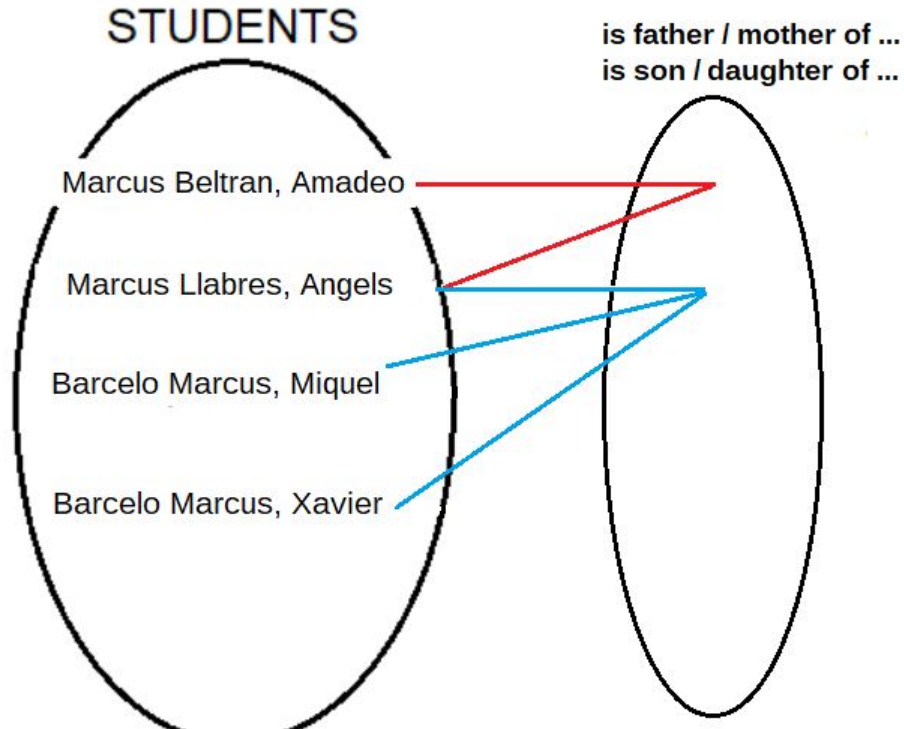
Relationships are ...

... associations, links or correspondences between items of entities related in some way in the real world.

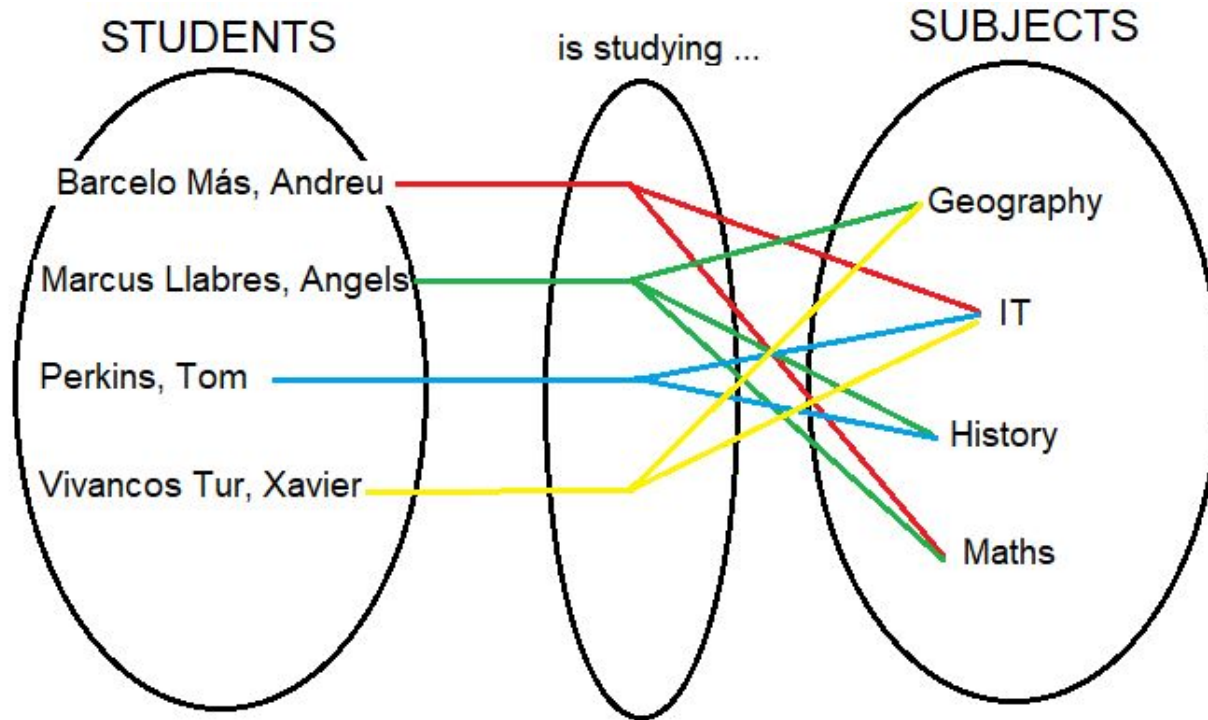
Must be translate as "*interrelació*" (cat) or "*interrelación*" (cast).



UD2 - Components of a E/R Diagram - Relationships (Reflexive)

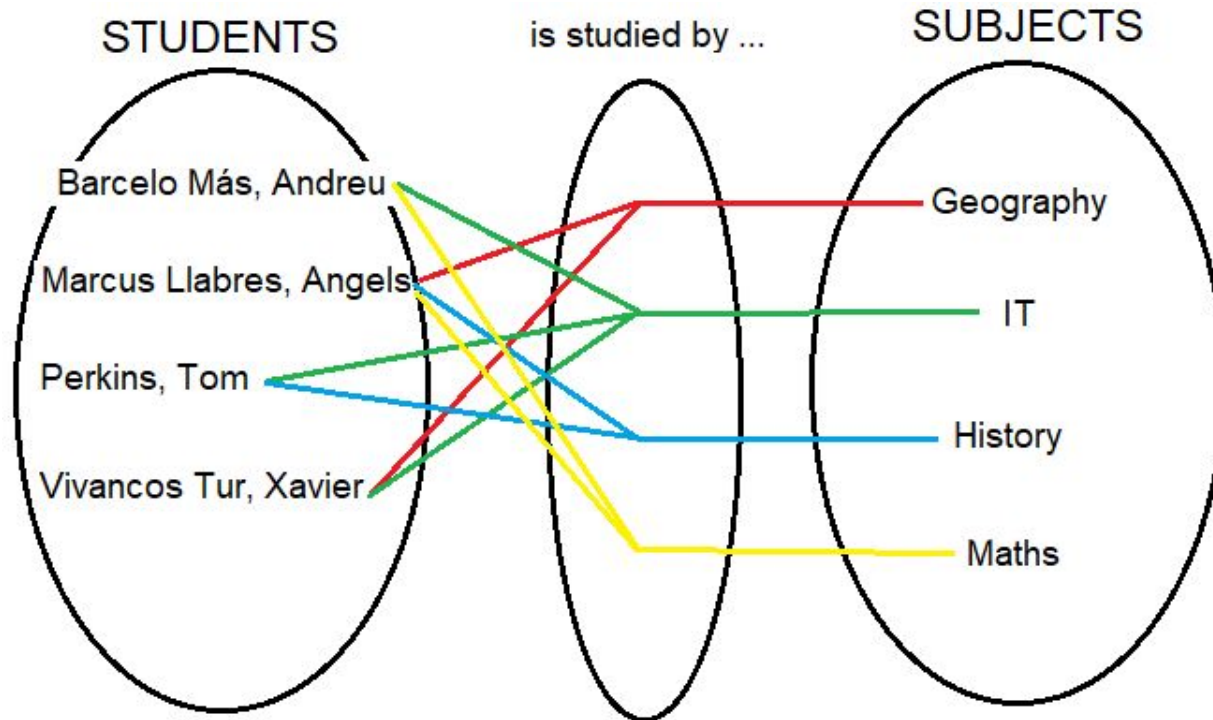


UD2 - Components of a E/R Diagram - Relationships - (Binaries)





UD2 - Components of a E/R Diagram - Relationships





UD2 - Components of a E/R Diagram - Relationships

Relationships have the following properties ...

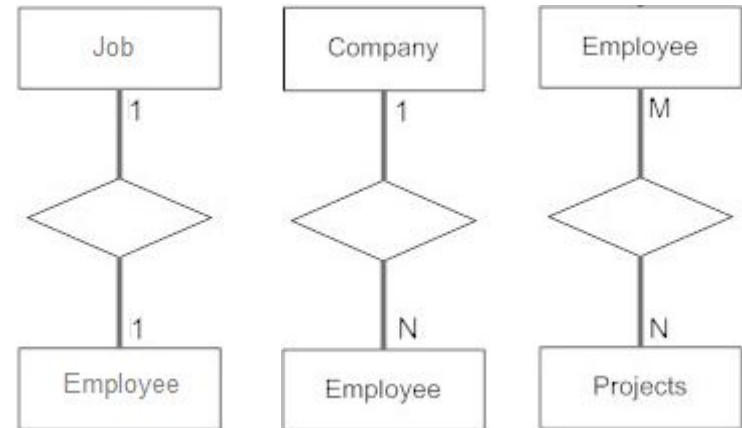
- **Name:** which provides an action to the relationship, therefore relationships are written as **verbs**.
- **Grade:** number of entities related in a relationship.
 - **Reflexive:** 1 entity is related on a relationship, They are used to represent hierarchical relationship between items on an entity.
 - **Binary:** most commonly used, 2 entities are related on a relationship. They are the most common relationship between entities.
 - **Ternary:** 3 entities are related: They may be treated as binary-relationships per pairs.



UD2 - Components of a E/R Diagram - Relationships

Relationships also have ...

- **Cardinality:** is the number of occurrences in one entity which are associated (or linked) to the number of occurrences in another. There are three degrees of relationship, known as:
 - one-to-one (1 : 1)
 - one-to-many (1 : N)
 - many-to-many (M : N)

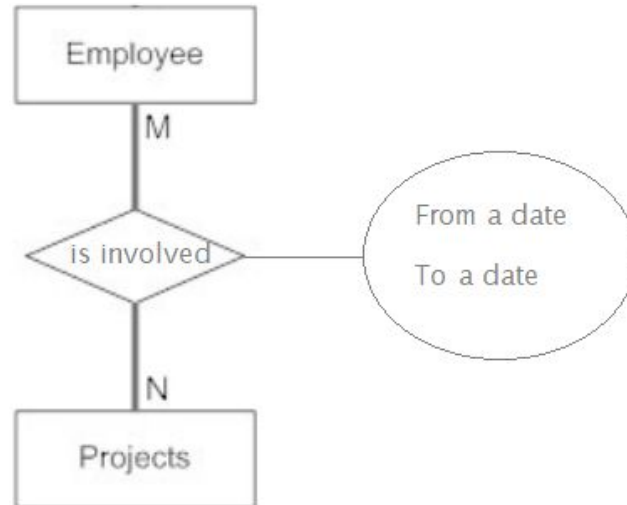




UD2 - Components of a E/R Diagram - Relationships

Relationships and may have ...

- **Attribute/s:** particularize the way in which a relationship between the instances are related.

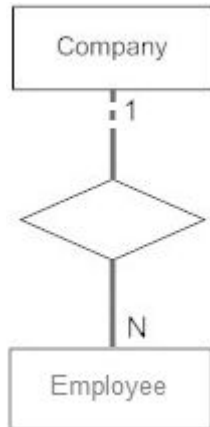




UD2 - Components of a E/R Diagram - Relationships

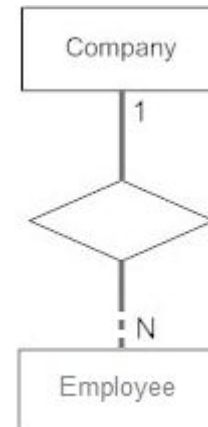
Relationships also have ...

- **Mandatory / Optionality:** a cardinality in a relationship may also be optional or mandatory. Either end of the relationship can include zero occurrences as an option or at least one as mandatory.



A company can have Employees related or have no one.

Common usage for a 1 : N relationship.



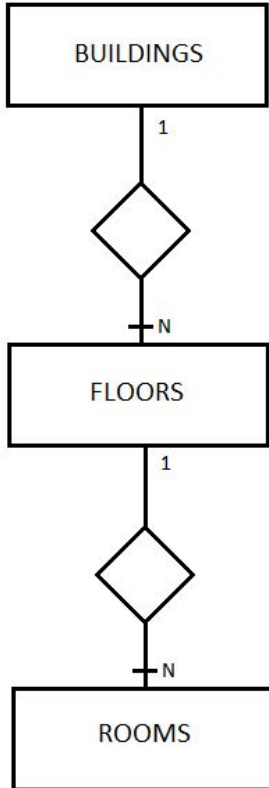
An employee can be related to a Company or to be related to no one Company.

Uncommon usage for a 1 : N relationship.



UD2 - Components of a E/R Diagram - Relationships

Strong and weak entities are ...



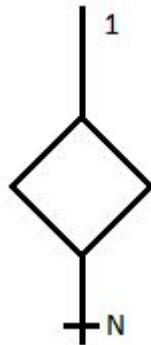
- **Strong entity:** is the one whose existence does not depend on the existence of any other entity in a schema.
 - A strong entity always has a primary key that describes the strong entity.
- **Weak entity:** is the one whose existence depends on a strong entity for its existence.
 - The primary key of a weak entity is a composite key formed from the primary key of the strong entity and the primary key of the weak entity.



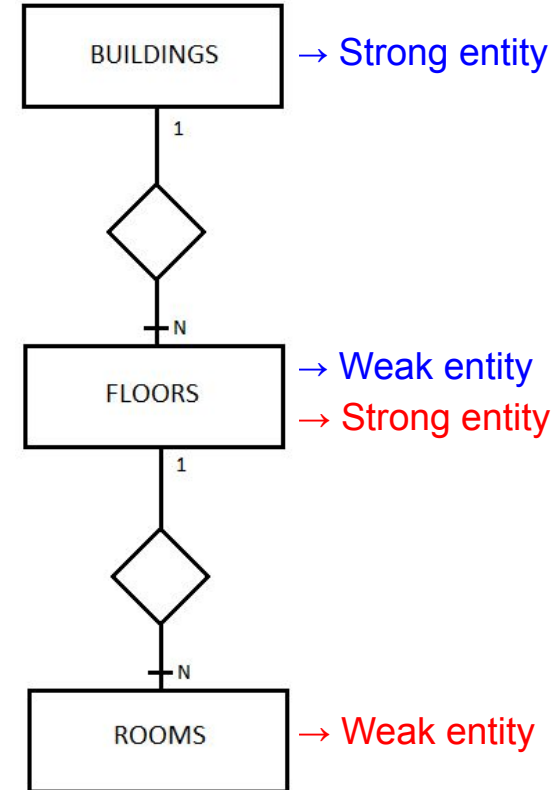
UD2 - Components of a E/R Diagram - Relationships

Relationships also have ...

- Transferable: related to relationships 1 : N and strong / weak entities.



→ Transferable relationship





UD2 - Components of a E/R Diagram - Relationships

Relationships are represented by ...

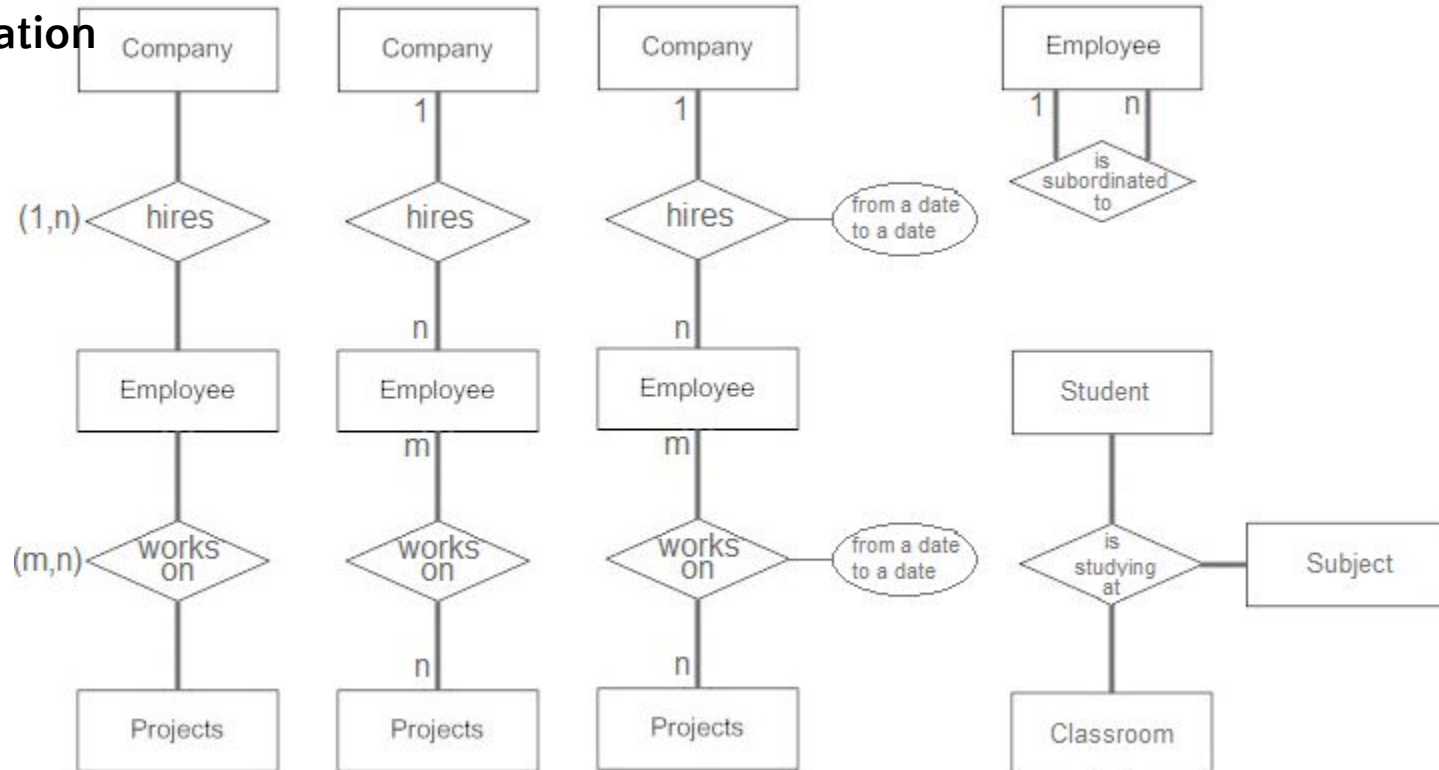
There are many representations of an ERD. Below are some examples:

- **Standard notation**
- **Depending on the author:**
 - Chen notation,
 - Bachman notation,
 - Martin notation,
 - Barker notation.



UD2 - Components of a E/R Diagram - Relationships

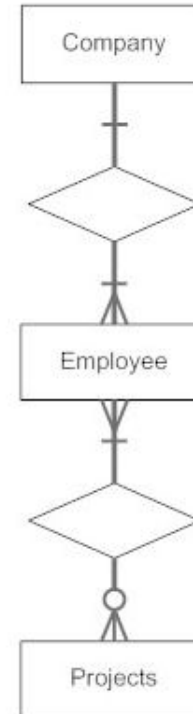
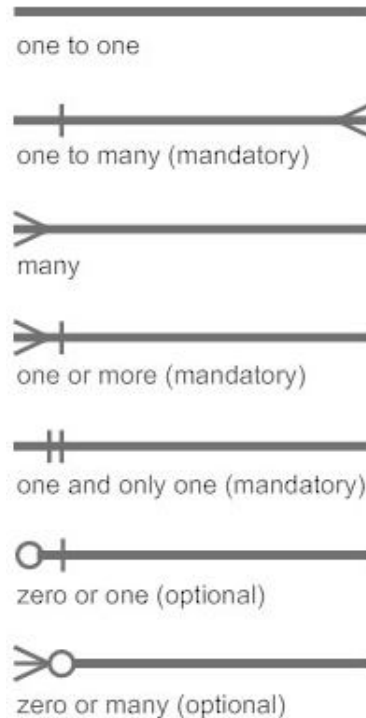
Standard notation





UD2 - Components of a E/R Diagram - Relationships

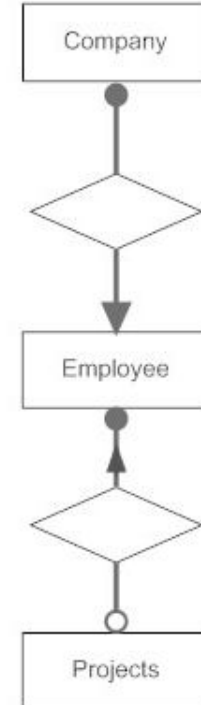
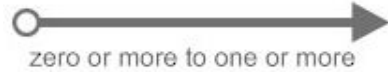
Chen notation





UD2 - Components of a E/R Diagram - Relationships

Bachman notation





UD2 - Components of a E/R Diagram - Relationships

Martin notation

1 - one, and only one (mandatory)

* - many (zero or more - optional)

1...* - one or more (mandatory)

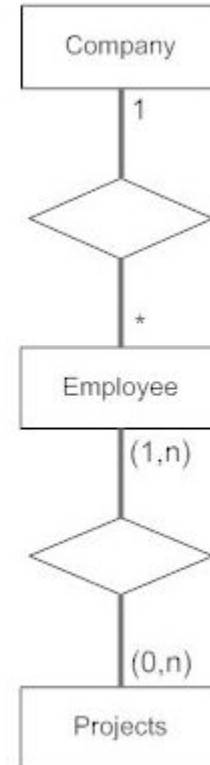
0...1 - zero or one (optional)

(0,1) - zero or one (optional)

(1,n) - one or more (mandatory)

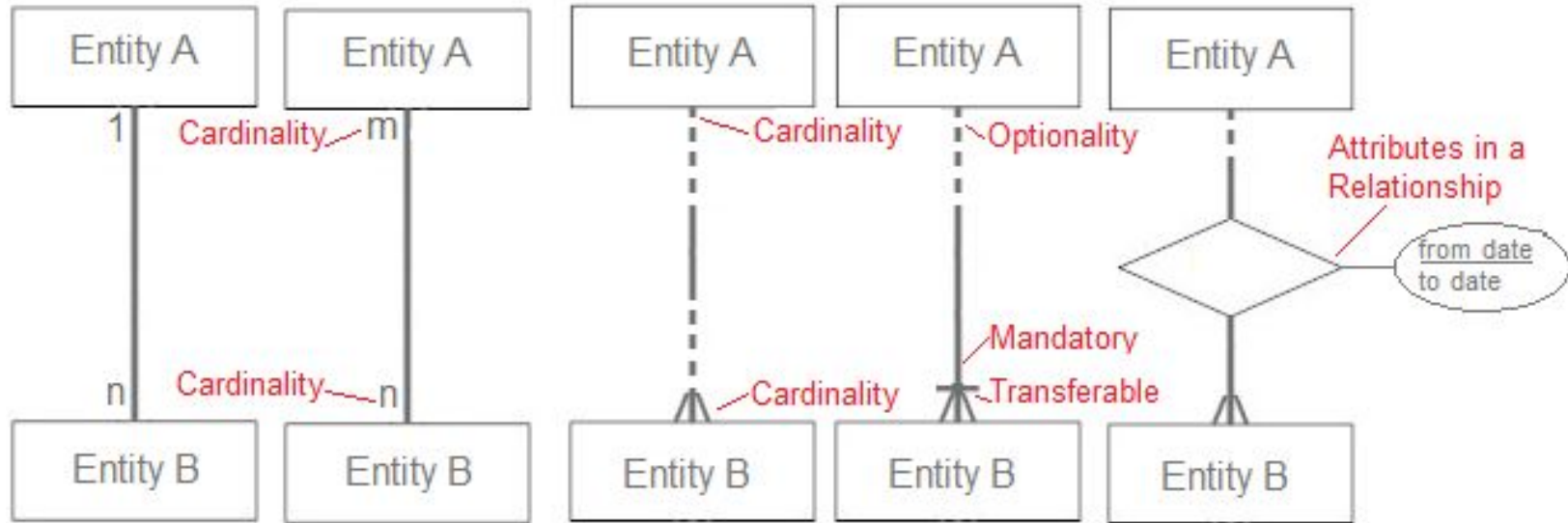
(0,n) - zero or more (optional)

(1,1) - one and only one (mandatory)





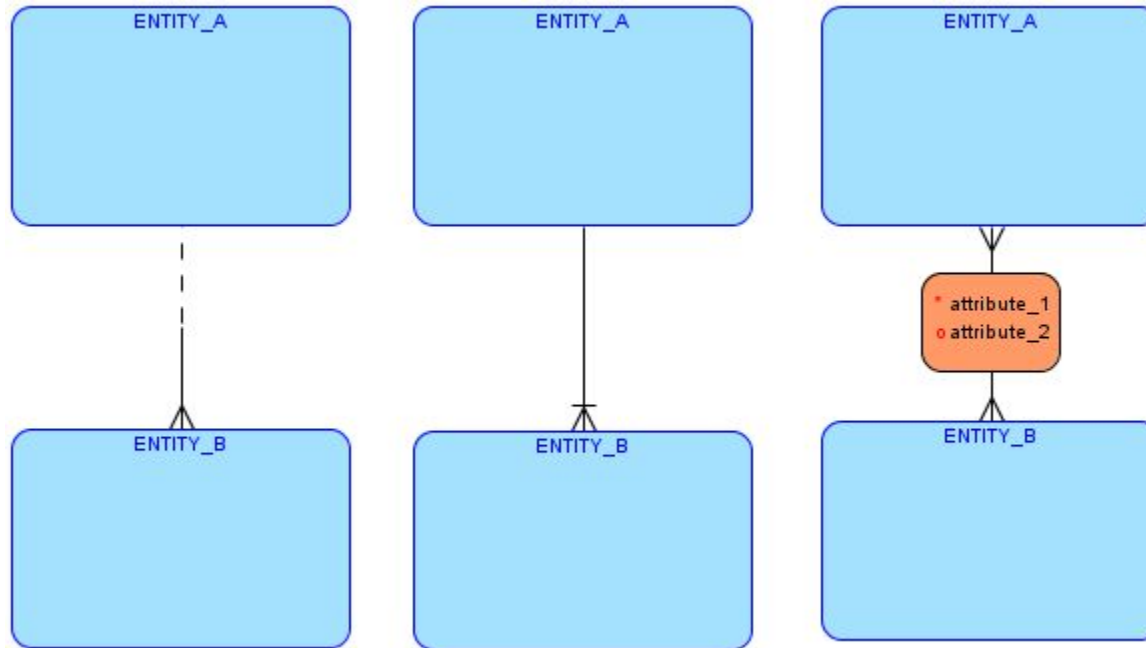
UD2 - Components of a E/R Diagram - Relationships





UD2 - Components of a E/R Diagram - Relationships

The notation we will use (Barker Notation) ...





UD2 - Components of a E/R Diagram - Domains

Domains are ...

... a set of values which an attribute can derive from. Each simple attribute is associated with a domain, which specifies its valid values.

A domain can be defined from different ways:

- **Extension:** defined from an assertion
- **Intention:** defined from the possible values it can take

Examples:

- NAMES: a character string of up to 30 chars.
- DESCRIPTIONS: a character string of up to 100 chars.
- PHONE-NUMBERS: a character string of up to 9 numbers.
- GENDERS: a character string of 1 char: M $\rightarrow$ Male, F $\rightarrow$ Female.



UD2 - Components of a E/R Diagram - Domains

Domains are represented by ...

It is not used to represent the relationship between domains and their attributes.

- In any case, it can be represented by a table.
- Domain names are plural and shown in uppercase.



UD2 - Components of a E/R Diagram - Domains

DOMAIN	ATTRIBUTES
NAMES	name, first-name, last-name
DESCRIPTIONS	full-name, subject-name, country-description, department-description, ...
...	...



UD2 - An E/R Model Extension - Generalization, Specialization, Aggregation

As part of the Enhanced ER Model, along with other improvements, three new concepts were added to the existing E/R Model, they were:

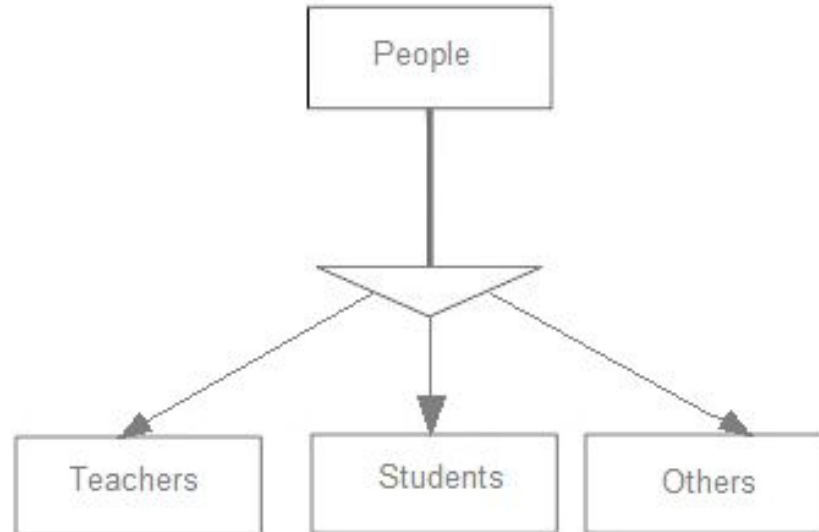
- Specialization,
- Generalization and
- Aggregation.



UD3 - An E/R Model Extension - Generalization, Specialization, Aggregation

Specialization is ...

When sub-entities are created from a Parent entity, it is called specialization.

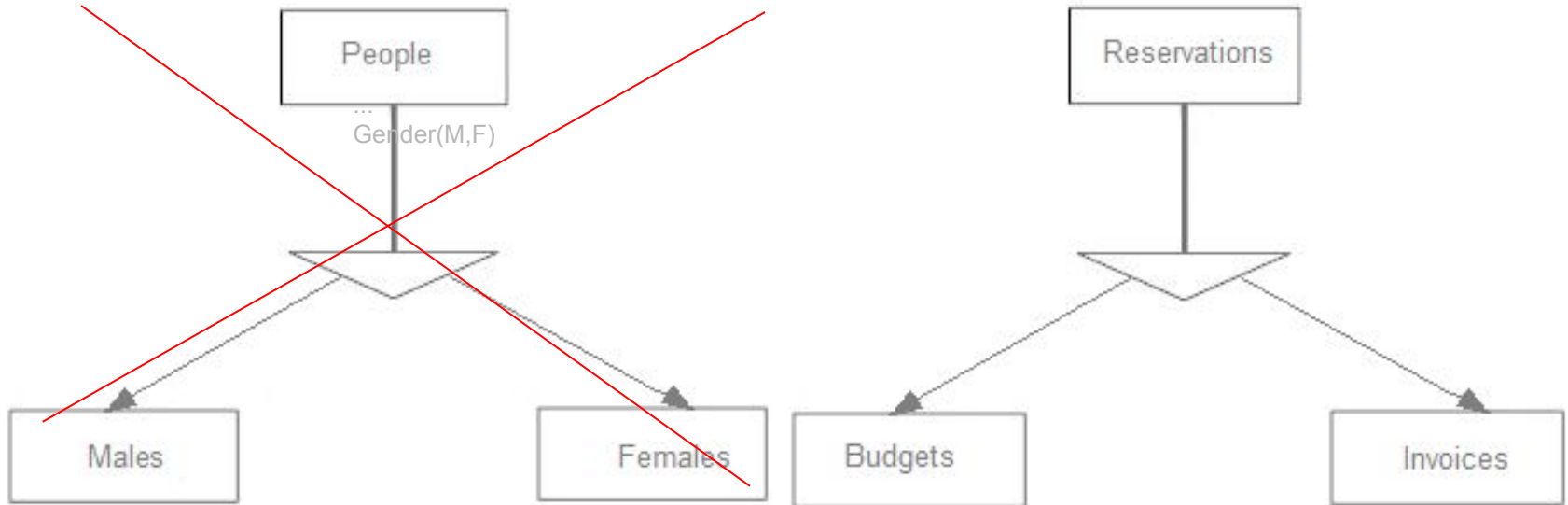




UD3 - An E/R Model Extension - Generalization, Specialization, Aggregation

Generalization is ...

When certain number of sub-entities "Daughters" are committed to create an superentity "Father".





UD3 - An E/R Model Extension - Generalization, Specialization, Aggregation

Transformation rules for Generalization, Specialization, Aggregation are ...

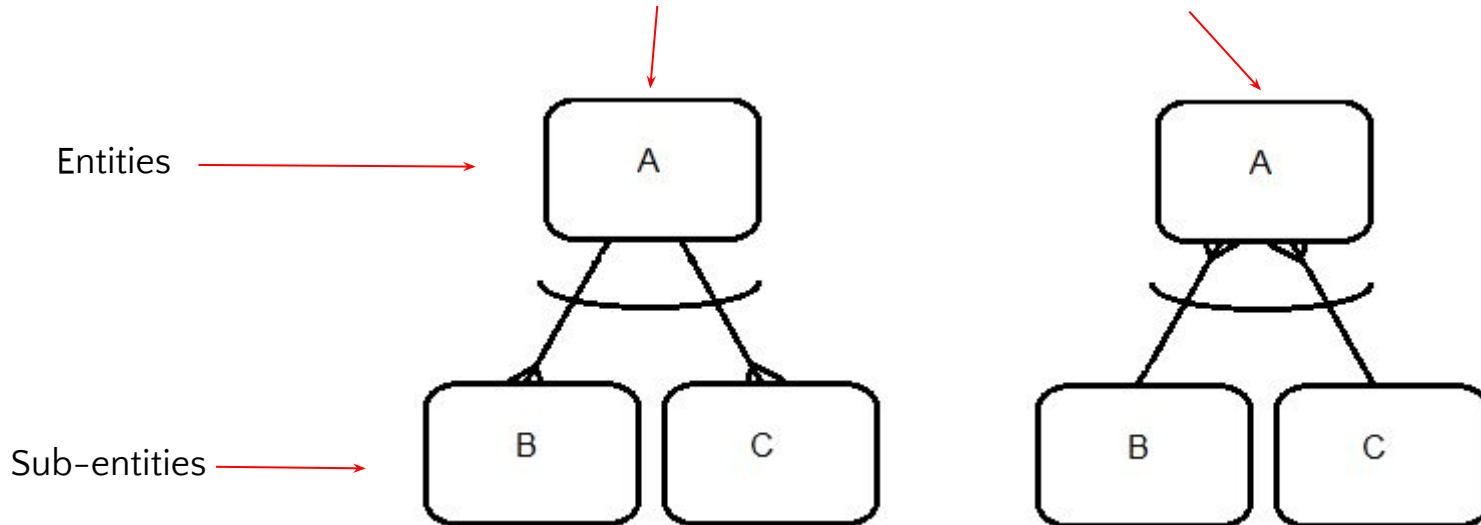
- Specialization and Generalization are treated as Arc-Relationships.
- There is no transformation to Relational Model for Aggregations so that they will not be treated at the moment.



UD3 - An E/R Model Extension - Arc Relationship Transformation

Arc-Relationships provide ...

... a proper specialization OR a proper generalization treatment.





UD2 - Introduction to UML

UML (Unified Modeling Language) is a standardized modeling language consisting of an integrated set of diagrams useful in Software Engineering.

Examples of UML Diagrams are:

- Case of use Diagram,
- Class Diagram,
- Flux Diagram,
- ...



UD2 - Introduction to UML

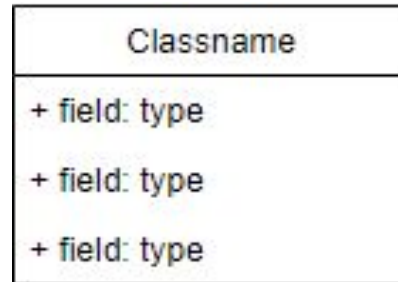
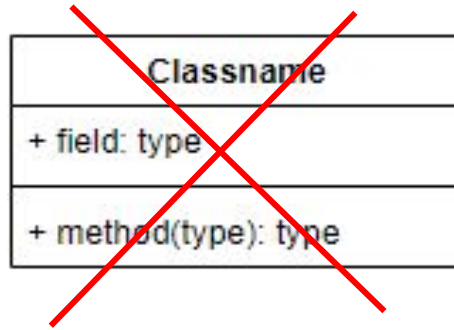
In the context of Oriented Object Software Engineering, a **Class** is "*a set of objects that share the same attributes, operations, methods, relationships, and semantics*". **Certainly an Entity and a Class have the same definition ...**

A **Class Diagram** describe all the components of a Class in a simple way.

In this way, it is used to represent E/R diagrams by using a Class Diagrams.

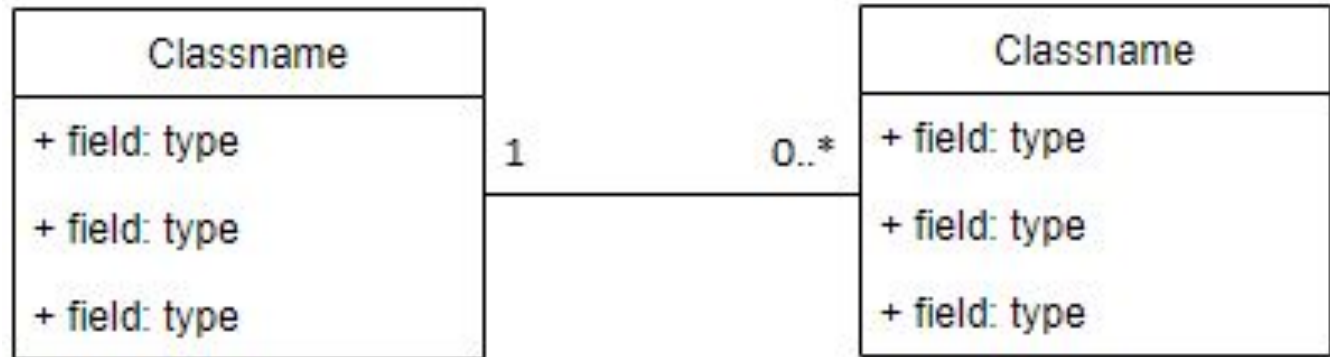


UD2 - Class Diagram - Classes



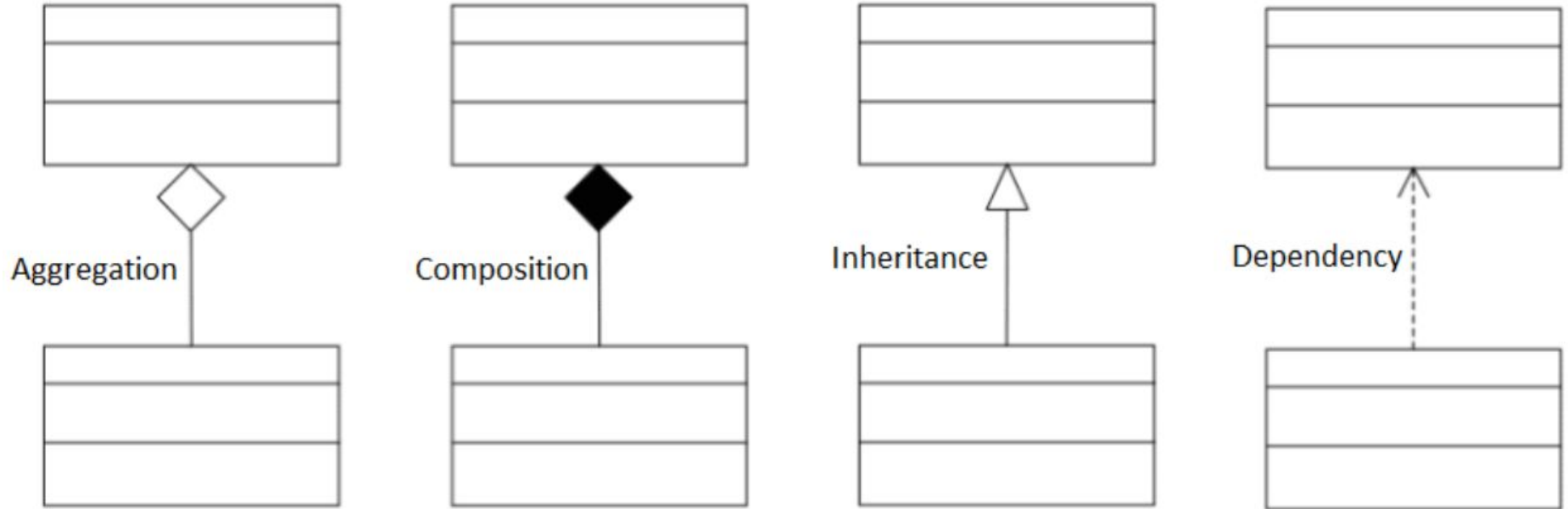


UD2 - Class Diagram - Classes and their Relationships





UD2 - Class Diagram - Type of Relationships





UD2 - Class Diagram - Type of Relationships



1. **Aggregation:** Define Objects that “are-part-of” another complete object. The destruction of the compound does not entail the destruction of the components. It usually occurs more frequently than the composition.



2. **Composition:** Define a strong form of aggregation where the lifetime of the contained class must match the lifetime of the containing class.



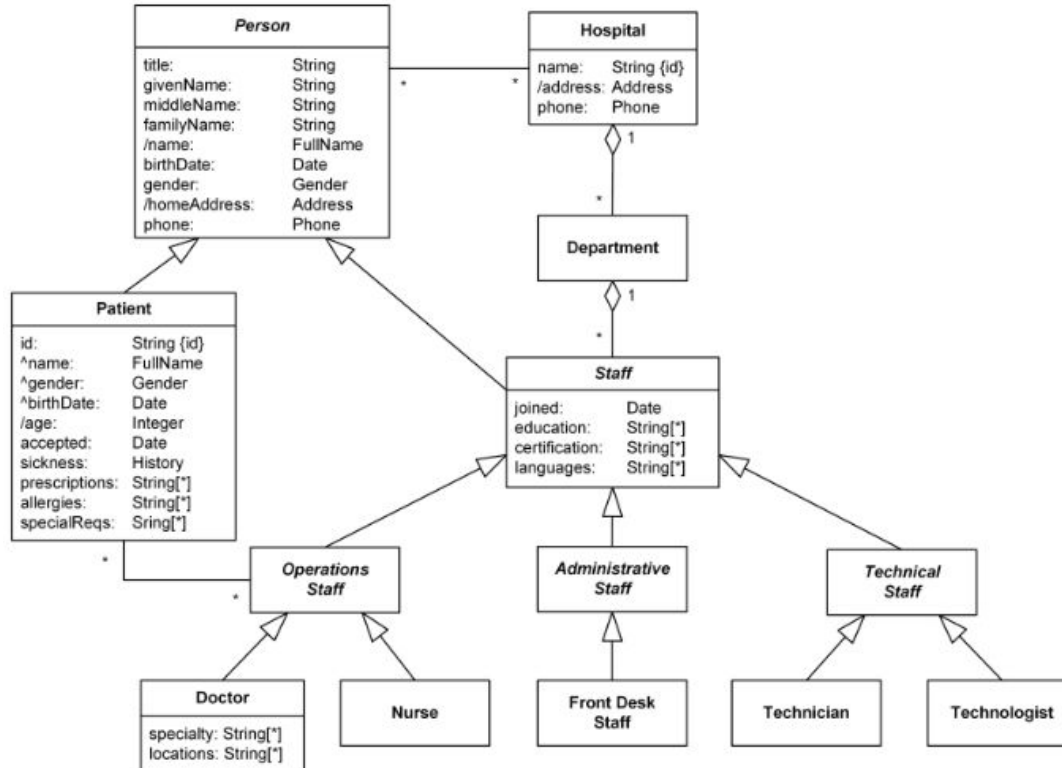
1. **Inheritance:** *Hierarchies of generalization/specialization are known as inheritance.* Inheritance is the mechanism that allows a class of objects to incorporate attributes and methods of another class, adding them to those it already has. Inheritance reflects an "is_a" relationship between classes. The class from which it inherits is called a superclass, and the class that it inherits from is called a subclass.



2. **Dependency:** related to Master and Detail Entities.

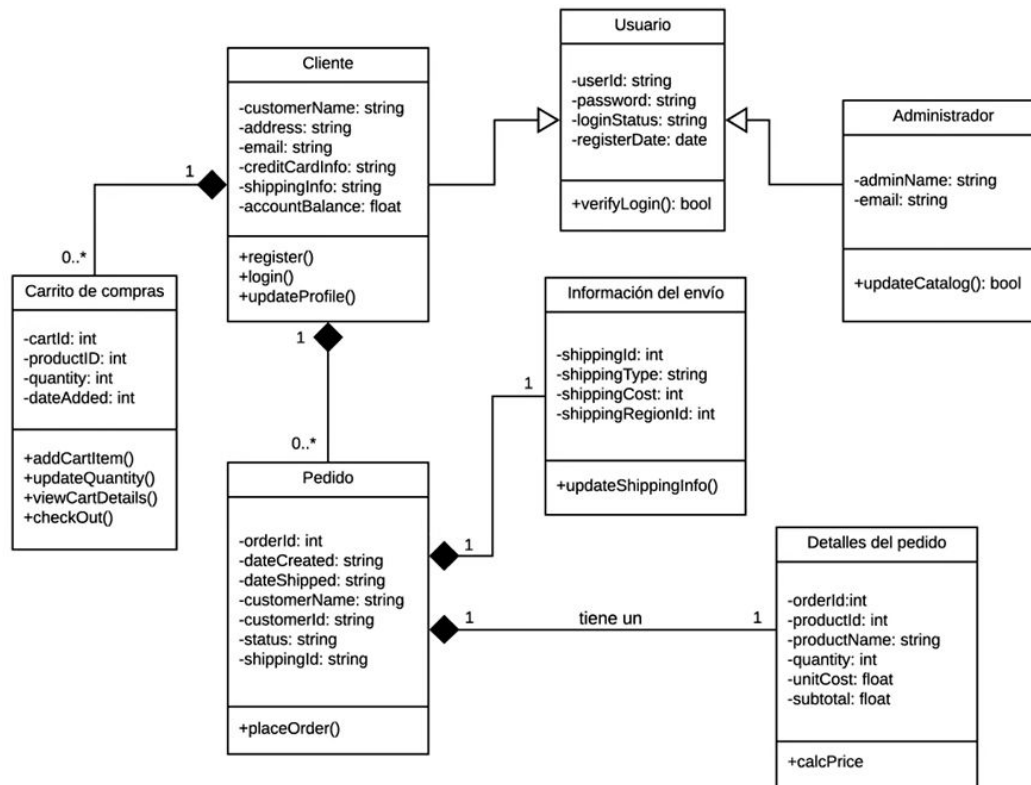


UD2 - Class Diagram - Example





UD2 - Class Diagram - Example





Recursos de la UD

Additional resources

- [Oracle Data Modeler](#) - [User's guide](#), [Learn more ...](#)
 - [Tutorial: Del modelo lógico al modelo Físico](#)
- [SQL Workbench](#) - [EER Diagram Editor](#)
- [Enterprise Architect](#) - [Enterprise Architect](#)
- MS PROJECT

Assignments

- ACT2.1 - ERD Kick off
- ACT2.2 - ERD - Entities, their Attributes and Relationships
- ACT2.3 - ERD - Multiple Relationships
- ACT2.4 - ERD of a Library
- ACT2.5 - ERD of a Geography Location

- ACT2.6 - ERD of an Academy
- ACT2.7 - ERD of a Hospital
- ACT2.8 - ERD of an Academy
- ACT2.9 - ERD of Film Festivals
- ACT2.10 - Reverse ERD
- ACT2.11 - Practice test UD2
- ACT2.12 - ERD of a Chess Championship
- ACT2.13 - ERD of Energy Treatment
- ACT2.14 - ERD of The Management of a Tournament



Bibliografía

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 - <https://www.youtube.com/watch?v=MRmmPJld5-k>
 - <https://www.youtube.com/watch?v=u2bXiPJf9oQ>
 - <https://www.youtube.com/watch?v=te-i37IIFeU>

- Relaciones n-árias
 - [Cardinalidad de cada relación n-ária](#)
 - [Transformación relaciones n-àrias](#)



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Adaptar — remesclar, transformar i crear a partir del material

El llicenciador no pot revocar aquestes llibertats, sempre que segueixi els termes de la llicència.

Amb els termes següents:

Reconeixement — Heu de reconèixer l'autoría de manera apropiada, proporcionar un enllaç a la llicència i indicar si heu fet algun canvi. Podeu fer-ho de qualsevol manera raonable, però no d'una manera que suggereixi que el llicenciador us dona suport o patrocina l'ús que en feu.

NoComercial — No podeu utilitzar el material per a finalitats comercials.

CompartirIgual — Si remescleu, transformeu o creeu a partir del material, heu de difondre les vostres creacions amb la mateixa llicència que l'obra original.

No hi ha cap restricció addicional — No podeu aplicar termes legals ni mesures tecnològiques que restringeixin legalment a altres de fer qualsevol cosa que la llicència permet

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